Indian Institute of Technology Palakkad

Algorithms

Assignment: Asymptotics

1. Is $\sqrt[10]{n}$ asymptotically larger than $(\log n)^{10}$?
2. Express $n^n$ and $2^{n^2}$ as exponential functions with same base and then compare them asymptotically.
3. Show that if $\log f(n)$ is in $\Theta(g(n))$, then $2^{f(n)}$ is not necessarily in $\Theta(2^{g(n)})$.
4. Prove that following asymptotic relations.
   (a) $2^n + n^2$ is in $o(3^n)$.
   (b) $2^n + n^2$ is in $\Theta(2^n)$.
   (c) $2n^2 + n\log n + 1$ is in $\Theta(n^2)$.
   (d) $n\sqrt{n} + n\log n + 2$ is in $\Theta(n\sqrt{n})$.
   (e) $2^{n+1}$ is in $\mathcal{O}(2^n)$.
   (f) $2^{2n}$ is not in $\mathcal{O}(2^n)$.
   (g) If $\log f(n)$ is in $\mathcal{O}(\log g(n))$, then $f(n)$ is in $\mathcal{O}(g(n))$.
   (h) $\log n!$ is in $\Theta(n \log n)$.
   (i) $n!$ is in $o(n^n)$ and in $\omega(2^n)$.
   (j) $n!$ is in $\Theta((\frac{n}{e})^n\sqrt{n})$.